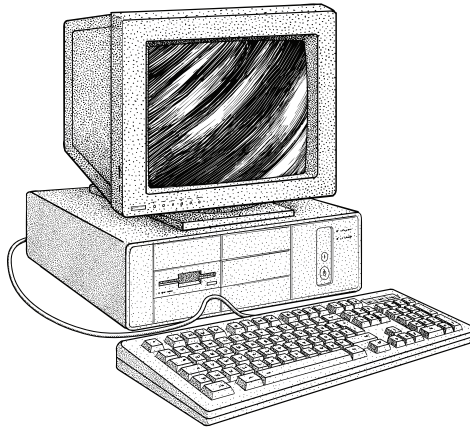


resents a significant departure from classical computation. Unlike a bit, a qubit can exist in a state of 0, 1, or both simultaneously, thanks to the principle of superposition. This ability allows a single qubit to perform multiple calculations at once.

Parallelism and Computing Power

The superposition principle is just the start. When multiple qubits entangle, a phenomenon where the state of one qubit can depend on the state of another, regardless of distance, the computational power of a quantum computer increases exponentially with each additional qubit. This exponential scaling is absent in classical computing, where adding more bits linearly increases computing



power. Thus, while a classical computer with n bits can only be in one of 2^n possible states at any time, an n -qubit quantum computer can be in all of those 2^n states simultaneously. This parallelism allows quantum computers to solve certain types of problems much more efficiently than classical computers.

For example, consider the task of finding a specific item in an unsorted database. A classical computer would need to check each item one by one, requiring, on average, $N/2$ checks for a database of N items. However, Grover's algorithm, a quantum algorithm, can find the item in roughly $\sqrt{N}$ operations, showcasing quantum computing's potential for speedup in search problems.

DIFFERENCE BETWEEN CLASSICAL VS QUANTUM COMPUTING

Feature	Classical Computing	Quantum Computing
Basic Unit	Bit (binary digit), which can be either 0 or 1.	Qubit (quantum bit), which can exist in a state of 0, 1, or both simultaneously due to superposition.
Data Processing	Processes information in a binary format, executing operations sequentially.	Leverages superposition and entanglement to process multiple possibilities simultaneously.
Parallelism	Limited parallel processing capabilities, constrained by hardware.	Inherent parallelism through superposition, allowing exponential scaling with the number of qubits.
Computation Method	Utilizes classical logic gates to perform computations.	Uses quantum gates that manipulate qubits, enabling complex computations that involve probability.
Scalability	Linear scaling; doubling the number of bits doubles the processing power.	Exponential scaling; adding qubits exponentially increases the computational power due to the entanglement.
Error Rate	Relatively low error rates, errors can be easily detected and corrected.	Higher error rates due to qubit sensitivity; quantum error correction is more complex and still under development.
Main Applications	General-purpose computing, including business applications, scientific research, and personal use.	Specialized tasks like quantum simulation, cryptography, and optimization problems unsolvable by classical means.
Current State	Mature and widely used technology with extensive infrastructure.	Still in the developmental stage, with limited but rapidly advancing practical applications.
Security	Security based on complex algorithms that can be solved in reasonable time by powerful classical computers.	Potential to break current encryption methods using algorithms like Shor's, also introduces quantum encryption methods like QKD (Quantum Key Distribution) that are theoretically unbreakable.
Impact on Society	Pervasive impact across all sectors; the foundation of modern digital society.	Emerging impact, promising revolutionary advances in fields such as medicine, cryptography, and materials science.